Face Recognition System

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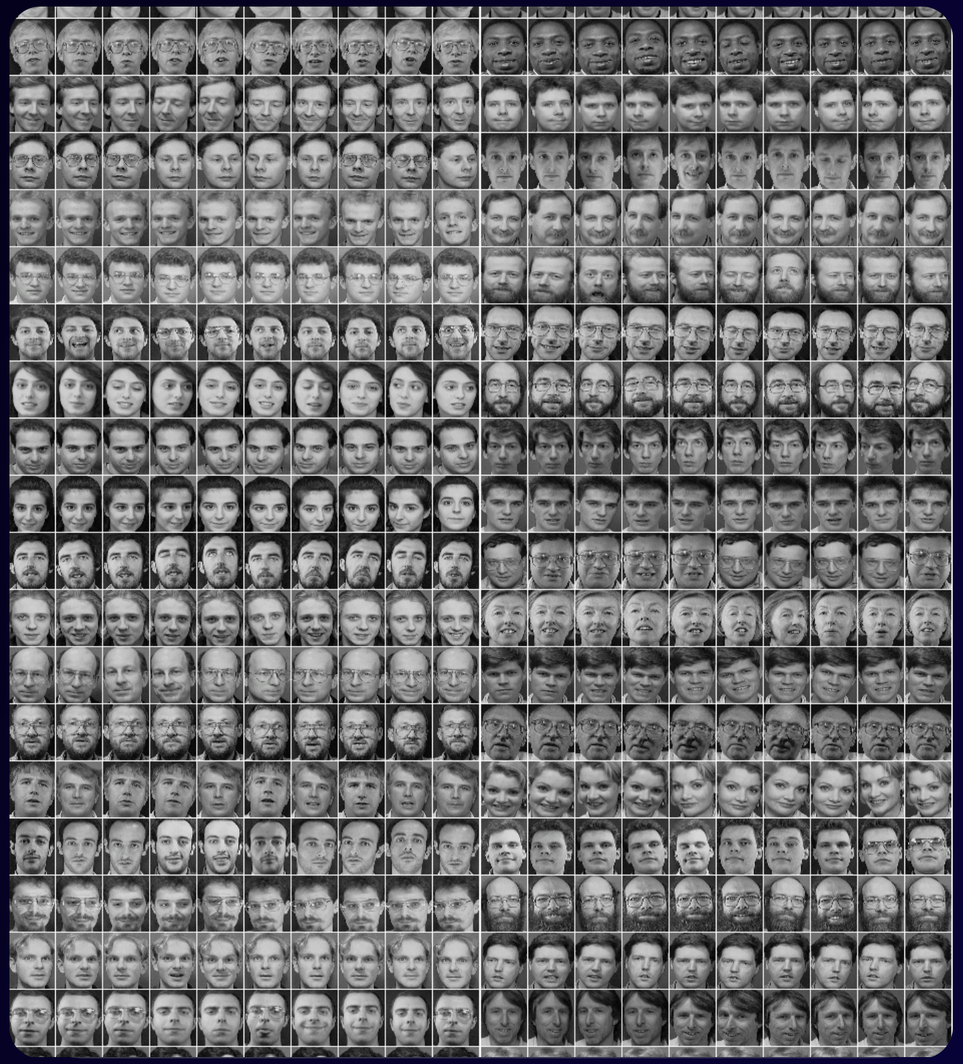
Project Overview

A comprehensive machine learning system that implements and compares multiple face recognition algorithms to identify the most effective approach for facial recognition tasks.

Dataset

- Source: AT&T Face Database (Olivetti Faces)
- Size: 400 images total
- Subjects: 40 different people
- Images per person: 10 photos each
- Image dimensions: 64×64 pixels (grayscale)
- Split: 80% training, 20% testing



Implemented Algorithms

1. Support Vector Machine (SVM)

- Approach: Traditional machine learning classifier
- Preprocessing: Image flattening + standardization
- Kernel: RBF (Radial Basis Function)
- Use case: Baseline comparison, good for smaller datasets

2. Neural Network (MLP)

- Architecture: Multi-layer perceptron with 3 hidden layers (512→256→128 neurons)
- Activation: ReLU
- Optimizer: Adam
- Use case: Deep learning approach without convolutions

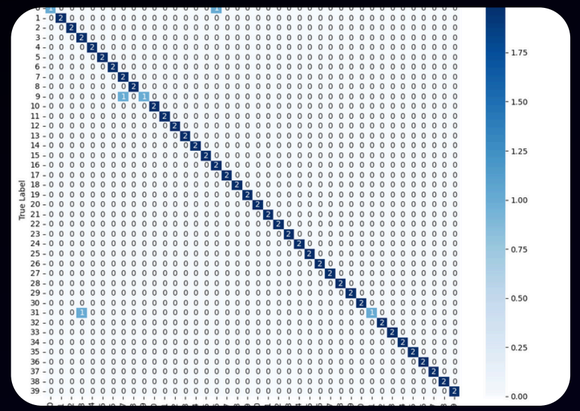
3. PCA + SVM

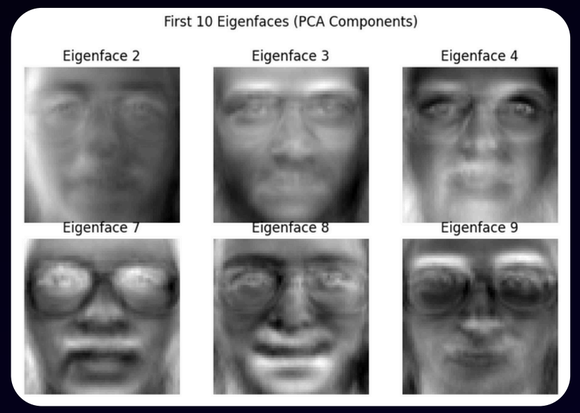
- Preprocessing: Principal Component Analysis for dimensionality reduction
- Components: 50-100 principal components
- Classifier: SVM on reduced features
- Advantage: Faster training, noise reduction
- Bonus: Generates eigenfaces visualization

4. K-Means Clustering

- Type: Unsupervised learning
- Preprocessing: PCA for dimensionality reduction
- Clusters: 40 (matching number of people)
- Evaluation: Cluster-to-label mapping based on majority vote

Project Key Features





1. Model Training & Evaluation

- Stratified train/test split to ensure balanced representation
- Comprehensive accuracy metrics (training and testing)
- Confusion matrices for detailed error analysis
- Classification reports with precision, recall, and F1-scores

2. Visualization & Analysis

- Model accuracy comparison charts
- Training loss curves for neural network
- Confusion matrices for each algorithm
- Eigenfaces visualization (first 10 principal components)

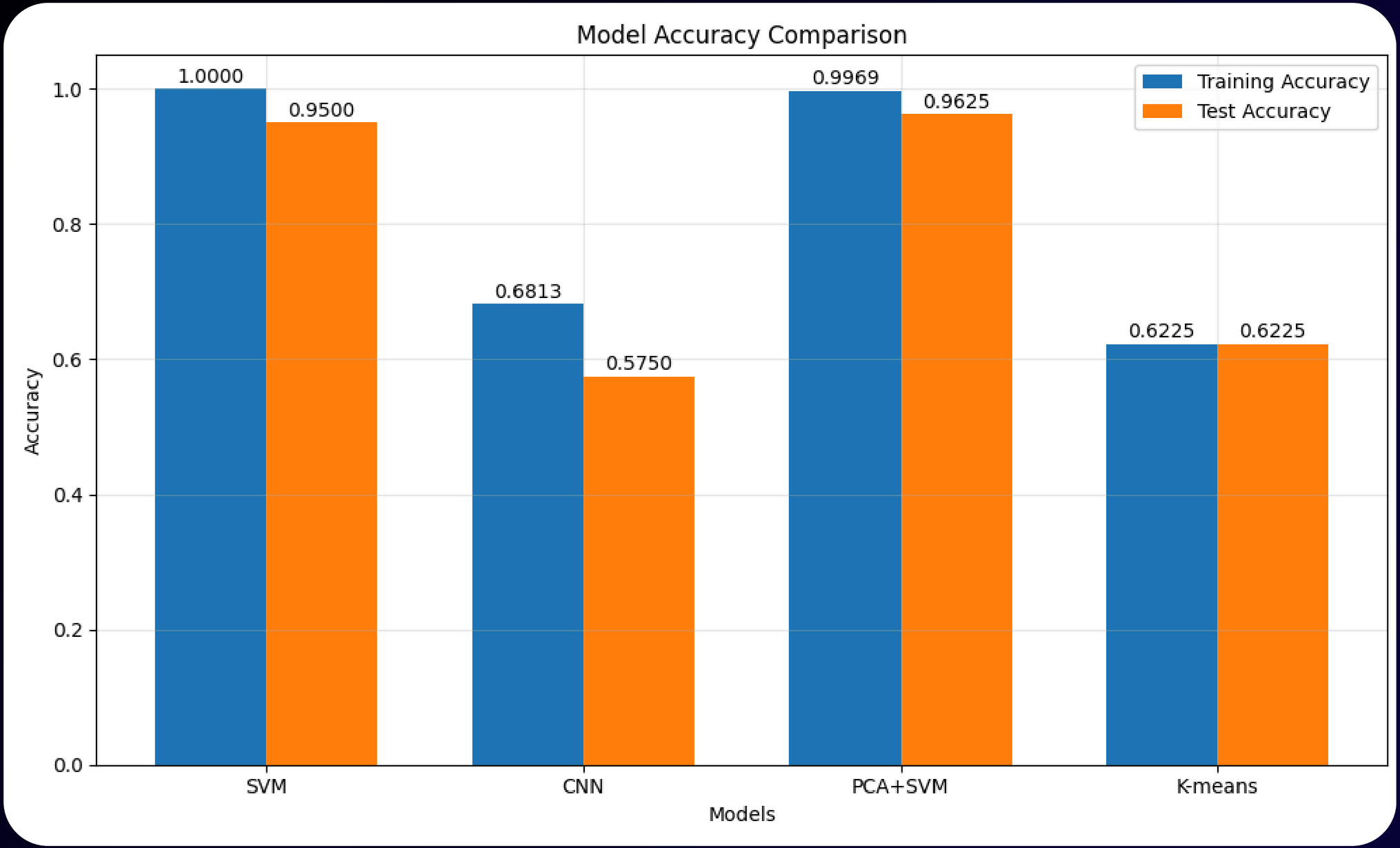
3. Prediction System

- Real-time prediction on new face images
- Confidence scores for predictions
- Support for multiple model backends
- Image preprocessing pipeline (resize, normalize)

4. Model Persistence

- Save/load trained models using pickle
- Organized model storage in dedicated folders
- Scalers and preprocessors included in saved models

Accuracy Comparison



Let's implement and demonstrate both Tkinter and Streamlit interfaces.

Thank You